

(a) Event fidelity

Complete event trace



Every event keeps its summary strip.

low = summary

high = summary

+ archived image

PROMOTE
attach image

DEMOTE
keep summary

Cold visual archive



Archive pixels are reattached.
Never text-generated.

(b) Fixed-B reallocation

Recent-4

B = 4



Promote the latest B events

CausalCache e3 replaces e5; B remains 4



LOCKED INVARIANTS

same goal + complete summaries
same current screen
same resolution + B active images
only promoted events change

Promote e3 only if $\Delta_t(e3 | S) > \text{utility of displaced } e5$
k is realized, not preset
B is not archival capacity

(c) Learning & deployment

Training: matched B;
only restoration identity changes



Other tokens bypass HGKV
B=0: exact bypass

HGKV
history pixels

Deployment

CausalCache-P



P: proposal-conditioned default

CausalCache-LA



LA: efficiency branch, not default